CHAPTER 5 · CONCEPT 3 · ANSWER KEY

Digits in Disguise — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1	**(a) a single digit (same letter = same digit).**
Q2	**(c) 0 or 5.** Divisible by 5 means last digit 0 or 5.
Q3	**1.** 37 × 3 = 111, so C = 1.
Q4	**(a) 0.** 9 + z must be a multiple of 9; the smallest digit is z = 0.
Q5	**True.** That is the basic rule of a cryptarithm.
Q6	**(a) Both true, R explains A.** 6 + a + 2 = 8 + a; the digit-sum test is exactly the reason.

Short answer

Q7	**a = 1, 4 or 7.** 8 + a must be a multiple of 3.
Q8	**A = 3, B = 7, C = 1.** 37 × 3 = 111 (since 111 = 3 × 37).
Q9	**b = 0 or 6.** Even (for 2) and digit sum 12 + b a multiple of 3 (for 3).

Long answer

Q10	**z = 0 or 9; the numbers are 3105 and 3195.** Digit sum 9 + z must be a multiple of 9.
Q11	**b = 0, 2, 4, 6 or 8.** Divisible by 4 ⇒ last two digits "b0" divisible by 4 ⇒ b even.
Q12	**One solution: 6 × 74 = 444 (A = 6, B = 7, C = 4, D = 4).** (Also 3 × 37 = 111.) Check: 6 × 74 = 444. ✓

Total: 21 marks.